1) UN $\quad \epsilon = 0.19 \quad \rho(UN) = 14.3\ g/cc \quad \phi = 4\times10^{14}\ n/cm^2 s \quad \sigma_f = 587\ b$

$Q = ?$

$M(UN) = 235 \times 0.19 + 238 \times 0.81 + 14 = 251.43\ g/mol$

$N(U^{235}) = 14.3 \frac{1\ mol}{251.43\ g} \frac{6.022\times10^{23}\ at}{1\ mol} \frac{1U}{1UN} \times 0.19 = 6.507\times10^{21}\ U^{235}/cc$

$Q = E_f \sigma_f \phi N^{235} = (200\times10^6\ eV)(1.602\times10^{-19} \frac{J}{eV})(587\times10^{-24}\ cm^2)(4\times10^{14} \frac{n}{cm^2 s})$
$\times (6.507\times10^{21})$

$\underline{Q = 489.6\ W/cc}$

$N_{UN}(U^{235}) = N_{UO_2}(U^{235})$

$6.507\times10^{21} = 10.97 \frac{6.022\times10^{23}}{x235 + (1-x)238 + 2\times16} \frac{1U}{1UO_2} x$

$6.507\times10^{21} = \frac{6.606\times10^{24} x}{270 - 3x}$

$9.85\times10^{-4} = \frac{x}{270-3x}$

$0.266 - 2.955\times10^{-3} x = x$

$\underline{x = 0.265}$

2) T @ clad midpoint

T @ r=0.1

T_{cool}: 540K

K_c: 0.18 W/m-K $\quad K_f$: 0.04 W/cm-K $\quad h_{cool}$: 1.5 W/cm²-K

Q = 350 W/cm³ $\quad R_f$ = 0.45 cm $\quad t_g$ = 80 μm $\quad t_c$ = 0.06 cm

$$T_{co} - T_{cool} = \frac{LHR}{2\pi R_f}\frac{1}{h_{cool}}$$

$$LHR = \pi R_f^2 Q = \pi (0.45)^2 350 = 222.66 \text{ W/cm}$$

$$\frac{LHR}{2\pi R_f} = \frac{222.66}{2\pi 0.45} = 78.75 \text{ W/cm}^2$$

$$T_{co} = 78.75\frac{1}{1.5} + 540\text{ K}$$

$$= 592.5\text{ K}$$

$$T_{c1} - T_{co} = \frac{LHR}{2\pi R_f}\frac{t_c}{K_c} = 78.75\frac{0.06}{0.18} = 26.25\text{ K}$$

$$T_{c1} = 618.25\text{ K}$$

$$T(\text{clad midpoint}) = \frac{T_{co} + T_{c1}}{2} = \frac{618.75 + 592.5}{2} = \boxed{605.6\text{ K}}$$

$$T_s - T_{c1} = \frac{LHR}{2\pi R_f}\frac{t_g}{K_g}$$

$$K_g = 16\times10^{-4}\,T^{0.79}$$

$$16\times10^{-4}(618.75)^{0.79} = 2.57\times10^{-3} \text{ W/cm-K}$$

$$T_s = 78.75\frac{80\times10^{-4}}{2.57\times10^{-3}} + 618.25$$

$$T_s = 863.9\text{ K}$$

$$T(r) = \frac{Q(R^2 - r^2)}{4k} + T_s = \frac{350(0.45^2 - 0.1^2)}{4(0.04)} + 863.9\text{ K}$$

$$\boxed{T(r=0.1) = 1285\text{ K}}$$

3) $L = 3.6\ m$ $\quad z_0 = 1.8\ m$ $\quad LHR^\circ = 400\ W/cm$ $\quad \gamma = 1.25$ $\quad T_{cool}^{in} = 580 K$

LHR @ $z = 2.1\ m$ $\quad T_{cool}^{out}$? $\quad C_p = 4200\ \frac{J}{kg \cdot K}$ $\quad \dot{m} = 0.1\ \frac{kg}{s \cdot rod}$

$$LHR\left(\frac{z}{z_0}\right) = LHR^\circ \cos\left(\frac{\pi}{2\gamma}\left(\frac{z}{z_0} - 1\right)\right)$$

$$LHR(z = 2.1) = 400 \cos\left(\frac{\pi}{2(1.25)}\left(\frac{2.1}{1.8} - 1\right)\right) = \boxed{391.3\ W/cm}$$

$$T_{cool}(z) - T_{cool}^{in} = \frac{2\gamma}{\pi}\frac{z_0\, LHR^\circ}{\dot{m}\, C_p}\left(\sin\frac{\pi}{2\gamma} + \sin\left[\frac{\pi}{2\gamma}\left(\frac{z}{z_0} - 1\right)\right]\right)$$

$$T_{cool}(3.6) = \frac{2(1.25)}{\pi}\frac{180\ (400)}{0.1\ (4200)}\left(\sin\frac{\pi}{2 \cdot 1.25} + \sin\left[\frac{\pi}{2 \cdot 1.25}\left(\frac{3.6}{1.8} - 1\right)\right]\right) + 580 K$$

$$= 259.5 + 580 = \boxed{839.5 K}$$

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4) FIMA = 5% $\quad K_{ox} = \frac{1}{A + BT}$ $\quad A = 3.8 + 200\ FIMA$ $\quad B = 0.0217$ $\quad T = 1200 K$

K_{ox} @ FIMA = 0

$$K_{ox} = \frac{1}{3.8 + 0.0217(1200)} = 0.0335\ W/cm\text{-}K$$

K_{ox} @ FIMA = 0.05

$$K_{ox} = \frac{1}{3.8 + 200(0.05) + 0.0217(1200)} = 0.025\ W/cm\text{-}K$$

$\boxed{\Delta K_{ox} = 0.0085\ \frac{W}{cm\text{-}K}}$ ~25% reduction

5. (6 pts) What assumptions to the heat conduction equation were required to derive the analytical form of the temperature profile?

Steady-state. Axisymmetric. No axial dependence. Constant material properties (no T or spatial variation)

6. (5 pts) What compound is utilized in the enrichment process? Describe the centrifuge-based enrichment of U, including why it works.

UF_6. Spinning cylinder, have a feed of UF_6 entering the cylinder. Due to mass differences between U235 and U238, the U238-F6 will preferentially accumulate at the outside of the cylinder, leaving the center of the cylinder slightly enriched in U-235F6. Tails are removed from the outside of the cylinder, enriched stream removed from center of cylinder. Process is repeated to obtain target enrichment.

7. (8 pts) Discuss the departure from nucleate boiling ratio and the critical heat flux.

The CHF is the point on the boiling curve which marks a transition from nucleate boiling to film boiling, corresponding to a drop in the maximum heat flux that can be accommodated by the coolant. If the CHF is exceeded, the coolant can no longer transport all the heat generated, and the fuel/cladding temperature will increase dramatically. The DNBR is the ratio of the CHF to the heat flux in the hottest channel. The DNBR should be maintained at about 1.3.

8. (4 pts) What is smear density? Why is a reduced smear density necessary?

Volume of the volume divided by the available volume inside the cladding. Reduces to a ratio of the squared radii of the fuel and the inside of the cladding. Allows for swelling of the fuel and creep down of the cladding.

9. (4 pts) What are two sources of thermal conductivity degradation in UO_2?

Point defects, voids, bubbles, precipitates, grain boundaries, etc…

10. (6 pts) What is the role of cladding? Why is Zr suitable as a cladding material?

Cladding is the primary containment. Retains fission products, keeps the shape of the fuel, while being neutron transparent and allowing heat transfer to the coolant. Has low neutron cross section, high thermal conductivity, corrosion resistance, cheap, etc.

11. (4 pts) What does the “fuel system” consist of?

Fuel, gap, cladding, coolant

12. (4 pts) What are the three aspects that I define as constituting fuel performance?

Heat generation and transport. Ability to operate under normal conditions without outages. Performance under accident scenarios.

13. (6 pts) What are the three ways that space is discretized for numerical solutions? Name a strength or weakness of one of these types.

Finite difference, finite volume, finite element. Finite difference is limited to grid-like geometries. Many other correct answers for strength/weakness.

14. (5 pts) Name two positive and two negative aspects of UO2 fuel.

Positive: Single phase, high melting point, ease of fabrication, corrosion resistance, radiation resistance, etc.
Negative: low U loading, low thermal conductivity, brittle, sensitive to O/M, etc.

15. (4 pts) What is the conceptual difference between implicit and explicit time integration?

Either taking the current state to make guesses about a future state (explicit), or using information about both the current and future states to make guesses about a future state (implicit).